



温州肯恩大学
WENZHOUCHEAN UNIVERSITY

Course Information

Course Title: Integrative Application of Business Analytics

Course Number and Section: MGS 4701 W01

Campus Location: Wenzhou-Kean University

Semester: 2026 Fall (Aug 31 – Dec 17)

Meeting Days and Times: Tue & Thu 13:00 – 14:15

Meeting Location: CBPM A101

Instructor Information

Instructor Name: Dawei CHEN

Office Location: CBPM B214

Office Hours: Tue 16:00 – 17:15 | Wed 15:00 – 16:00 | Thu 14:15 – 17:00

Email: dchen@kean.edu

Course Description

The purpose of this course is to provide students with hands-on experience to apply and integrate the knowledge and skills learned from all Business Analytics core courses to tackle real business problems. This is a rigorous team-based and project-intensive course. Moreover, the course challenges students to originate innovative solutions from strategic and technical perspectives.

Requisites

- MGS 3501 Database Management
- MGS 3701 Data Mining in Business

Course Level Student Learning Outcomes

Upon successful completion of this course, the student will be able to:

- A. Define and formulate a real business problem, or an area of improvement that may be developed using business analytics tools and techniques

- B. Evaluate the changes required on the strategic level to leverage data science and demonstrate the paradigm shift towards applying business analytics techniques
- C. Apply data analysis and data mining techniques to identify information and insights from huge volume of structured and unstructured data
- D. Synthesize the learning experiences acquired in the Business Analytics option to effectively recommend solutions for business problems
- E. Create standards and measures to track and evaluate the impact of the proposed solution on the organization

Section Level Student Learning Outcomes

Upon successful completion of **this section**, students will be able to:

- a. Design and execute a Python pipeline to collect business data from web sources
- b. Transform unstructured text into analysis-ready variables
- c. Select and justify a diagnostic, causal, or prescriptive method appropriate to a business question
- d. Integrate generative AI into an analytics workflow with explicit verification, and disclose its use to professional and academic standards
- e. Communicate analytical work to non-technical audiences through a written report, a dashboard, an oral defense, and a public code repository

Instructional Method

The course is delivered through lectures, hands-on labs, team consultations, and a semester-long group project. Lectures introduce concepts and methods; labs apply them in Python to each team's own data; consultations are checkpoints at which a team's progress is reviewed and its next stage approved. Two mock interview sessions provide structured practice for graduate-program, internship, and job interviews. All technical work uses Python (Jupyter Notebook in VS Code), with code managed in GitHub.

Course Materials

No textbook purchase is required. Slides, notebooks, readings, and datasets are distributed through Canvas.

Required: a laptop with Python 3, Jupyter, and VS Code installed, and a GitHub account.

Course Schedule

Date		Topic	Note
w1	09/01	Course introduction and house rules	Icebreaker, objectives, design principles, schedule, assessment, AI policy
	09/03	Group project introduction	Project tracks, deliverables, timeline, team formation plan drafting
w2	09/08	Web scraping I	Concepts and orientation
	09/10	Web scraping II	Collection via API, Requests, JSON, pagination
w3	09/15	Web scraping III	HTML scraping, BeautifulSoup
	09/17	Mock interview 1	Diagnostic round
w4	09/22	Web scraping IV	Selenium, wrap-up
	09/24	AI-assisted business analytics	Prompts → workflows → skills → agents; 4D AI fluency framework; generative AI across the analytics workflow
Assignment 1 – Project plan and data collection, DDL 09/28, 23:59			
w5	09/29	Consultation 1 – collection gate	Per team, 15 min. Scraper debugging
	10/01	National Day Holiday (No Class)	
w6	10/06		
	10/08	Unstructured data analysis I	From text to variables: extraction, discovery, measurement, TF-IDF and embeddings; rules, models, or LLMs
w7	10/13	Unstructured data analysis II	Discovery: semantic similarity, deduplication, clustering and topic modeling
	10/15	Unstructured data analysis III	Extraction and measurement: LLM-based extraction, classification and sentiment, annotation and validation
w8	10/20	Midterm exam review	
	10/22	Midterm exam	Closed-book, in class
w9	10/27	Lab 2: text analytics	Discovery, extraction and measurement on the team corpus
	10/29	Git and GitHub	Version control, repository structure, analytics portfolio
w10	11/03	Data visualization and dashboards	Audience-first design, building a dashboard from the team's own data
	11/05	Consultation 2 – dataset gate	Per team, 15 min. Cleaning, EDA, model-free analysis and visualization
Assignment 2 – Analysis-ready dataset and exploratory findings, DDL 11/04, 23:59			
w11	11/10	Diagnostic analytics	Metric change attribution, dimension splitting, base and mix effects, seasonality, external shocks
	11/12	Causal inference I	Framework, laboratory and field experiments, A/B testing

w12	11/17	Causal inference II	Observational data, DID, matching, PSM
	11/19	Lab 3: casual inference	Identification strategy applied to team data
w13	11/24	Prescriptive analytics I	Simulation, Monte Carlo, agent-based modeling
	11/26	Prescriptive analytics II	Optimization, linear programming
w14	12/01	Lab 4: prescriptive analytics	Formulating and solving a constrained decision problem
	12/03	Guest speaker	
w15	12/08	Consultation 3 – presentation gate	Per team, 15 mins
	12/10	Final presentation I	Teams 1–3, presentation + Q&A
w16	12/15	Final presentation II	Teams 4–5, presentation + Q&A
	12/17	Mock interview 2	Second interview round, cross-team synthesis of project findings
Assignment 3 – Final report, DDL 12/17, 23:59			

**The schedule is tentative and may be adjusted based on class progress.*

Course Assessment

Grade Breakdown

Individual	Participation	10%
	Midterm Exam	30%
Group	Assignments	10%, 10%, 20%
	Final Presentation	20%
Total		100%

1. Participation (10%)

- Students are expected to attend all class sessions and actively participate.
- Leave requests must be submitted via email (dchen@kean.edu) with appropriate supporting documentation.
- 1 point will be deducted for each unapproved absence.
- 3 consecutive unapproved absences or missing an exam will be reported to the University in accordance with academic policy.

2. Assignments (40%)

- 3 assignments serve as milestones for the group project (A1 and A2 are 10%, A3 is 20%). Please refer to the group project guidelines for details.
- Late submissions will receive a 50% deduction.

3. Midterm Exam (30%)

- Tentative date: 22nd October (tentative)
- Format: closed-book, in class

- No make-up exam will be provided.

4. Final Presentation (20%)

The presentation is a 30-minute defense in which every team member presents and answers questions on any part of the work. Please refer to the group project guidelines for details.

Grading Scale

Grade	Percentage Range	Quality Point	
A	≥ 93	4.0	Performance Excellent (A, A-) Good (B+, B-, B) Fair (C+, C) Poor (D) Fail (No Credit Granted)
A-	90 – 92.9	3.7	
B+	87 – 89.9	3.3	
B	83 – 86.9	3.0	
B-	80 – 82.9	2.7	
C+	77 – 79.9	2.3	
C	70 – 76.9	2.0	
D	60 – 69.9	1.0	
F	≤ 59.9	0.0	
IN	Incomplete		

AI Use Policy

Generative AI is a required tool in this course, not a prohibited one. Students may use AI for coding, data extraction, drafting, and revision, subject to three conditions:

- **Disclosure.** Every submitted deliverable includes an AI use statement specifying which tools were used, for what, and what the student changed.
- **Verification.** Students are accountable for the accuracy of all submitted work. “The AI produced it” is not a defense against an error.
- **Attribution.** AI-generated content presented as original student work violates Kean's Academic Integrity Policy.

Undisclosed AI use in graded work is treated as an academic integrity violation. AI use is not permitted during the midterm examination or in mock interview responses.

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